MCS-325	COMPILER TECHNIQUES	L T P 3 1 0	4 Credits

Course Objectives:

CO1	To discuss the techniques of scanning , parsing & semantic elaboration well enough to build or modify front end.
CO2	To expose the critical issues in modern compilers & provide them with the background to tackle those problems
CO3	Provide an understanding of the fundamental principles in compiler design
CO4	Learn the process of translating a modern high-level language to executable code required for compiler construction
CO5	Provide the skills needed for building compilers for various situations that one may encounter in a career in Computer Science.

MODULE I

Introduction: Definition , functions of Compiler in Linux / Unix / TC etc environments, other associated terms e.g. Text formatter, Text Editors, Phases and Passes, FSM & RE s and their application to Lexical Analysis, Implementation of Lexical Analyzers, Lexical- Analyzer Generator, Lex Compiler, Formal Grammar and their application to Syntax Analysis, BNF Notation, YACC.

The Syntactic specification of Languages: CFG, Derivation and Parse Trees, Capabilities of CFG.

Basic Parsing Techniques: Parsers, Shift Reduce Parsing, Operator precedence parsing, top down Parsing, Predictive Parsers.

MODULE II

Automatic Construction of efficient Parsers: LR Parsers, the canonical collection of LR(0) items, constructing SLR Parsing Tables, Constructing canonical LR Parsing tables and LALR parsing tables , An Automatic Parser Generator, Implementation of LR parsing Tables, Constructing LALR sets of items.syntax directed translation

Symbol Tables: Data Structure for Symbol Tables, representing scope information.

MODULE III

Run Time Administration: Implementation of simple Stack allocation scheme, storage allocation in block structured language.

Error detection and Recovery: Lexical phase errors,syntax phase errors,semantic errors Code Optimization: Loop optimization, the DAG representation of basic blocks, value numbers and Algebraic Laws, Global Data Flow Analysis.

Books and References:

1. Aho,Ullman & Sethi, Compiler Design , Addison Wesley
2 D.M.Dhamdhere, Compiler Construction Principles & Practice , Macmillan India Ltd.
3 Holub, Compiler Design in C , PHI.

Course Outcome:

CO1	Understand the basic concepts and application of Compiler Design
CO2	Apply their basic knowledge Data Structure to design Symbol Table, Lexical Analyzer
CO3	Understand and Implement a Parser -Top Down and Bottom Up Design
CO4	Understand strength of Grammar and Programming Language
CO5	Understand the concept of Code generation